A MULTIMODAL APPROACH TO ASPECT-BASED SENTIMENT ANALYSIS USING DEEP LEARNING

A PROJECT REPORT

Submitted in partial fulfilment of the requirements for the award of the degree of

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COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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CERTIFICATE

This is to certify that the project report entitled "**A Multimodal Approach to Aspect-Based Sentiment Analysis Using Deep Learning"** being submitted by **M. Satyamohan (22331A4237), M. Jagadeesh (22331A4219), T. Jeevan (22331A4257), and B. Rajesh (23335A4209)**, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning), is a record of bonafide work carried out by them under my supervision during the academic year 2025–2026.

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ABSTRACT

This paper introduces an end-to-end Aspect-Based Sentiment Analysis (ABSA) framework for Speech-to-Text Sentiment Analysis. Its core purpose is to convert the raw, unstructured speech from food vlogs into structured, actionable business intelligence. The proposed system is implemented as a lightweight browser extension. To address the challenges of transcription noise and the computational cost of full fine-tuning, the framework employs LoRA-based fine-tuning combined with Knowledge Distillation. This technique efficiently transfers the sophisticated Chain-of Thought reasoning capabilities of large language models into a more compact architecture. Furthermore, a novel dual-path Natural Language Processing (NLP) pipeline is utilized. This pipeline intelligently routes food-specific content to a specialized ABSA engine for aspect-level sentiment analysis, while using general sentiment analysis for non-food discussions. This enables the precise identification and sentiment classification of domain specific aspects—such as ambience, taste, and spiciness—even under noisy speech conditions. Experimental results confirm that the framework offers an efficient, scalable, and deployable solution. This empowers restaurant owners to make informed, data-driven decisions based on authentic, spoken customer feedback.

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List of Abbreviations

ABSA – Aspect-Based Sentiment Analysis

ASR – Automatic Speech Recognition

ASC – Aspect Sentiment Classification

ATE – Aspect Term Extraction

AI – Artificial Intelligence

ML – Machine Learning

NLP – Natural Language Processing

LLM – Large Language Model

BERT – Bidirectional Encoder Representations from Transformers

LSTM – Long Short-Term Memory

LoRA – Low-Rank Adaptation

TFN – Tensor Fusion Network

API – Application Programming Interface

GPU – Graphics Processing Unit

RAM – Random Access Memory

FFmpeg – Fast Forward Moving Picture Experts Group (Multimedia Framework)

KDD – Knowledge Discovery and Data Mining

EMNLP – Empirical Methods in Natural Language Processing

ACL – Association for Computational Linguistics

NAACL – North American Chapter of the Association for Computational Linguistics

ICASSP – International Conference on Acoustics, Speech, and Signal Processing

UGC – University Grants Commission

AICTE – All India Council for Technical Education

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CHAPTER 1

INTRODUCTION

In the contemporary digital era, the manner in which individuals communicate their opinions and experiences has undergone a profound transformation. Given the pervasive availability of the internet and mobile devices, social media platforms have emerged as principal conduits for expressing feedback concerning products, services, and personal encounters. Among these platforms, YouTube has become a notably influential repository of user-generated content, hosting millions of videos that encompass reviews, perspectives, and recommendations across diverse fields.

One of the most prevalent content formats on YouTube is verbal reviews, particularly within domains such as food critique, restaurant evaluations, travel commentaries, and service appraisals. In these videos, users articulate their experiences in comprehensive detail, frequently addressing multiple dimensions such as product quality, sensory attributes, setting, service efficiency, cost, and sanitation standards. These spoken reviews possess substantial engagement and persuasive power, as they more effectively convey affect, emphasis, and subjective impressions than written prose. Consequently, video-based reviews play a pivotal role in shaping consumer perceptions and influencing purchasing decisions.

Notwithstanding the richness and impact of verbal video reviews, the extraction of substantive insights from this medium remains a considerable undertaking. In contrast to textual reviews, which are inherently more structured and amenable to processing via conventional Natural Language Processing (NLP) methodologies, spoken content is intrinsically unstructured. Spoken discourse is conversational in nature and routinely incorporates disfluencies, hesitations, repetitions, elliptical sentence structures, and colloquialisms. Furthermore, ambient noise, regional accent variations, and articulatory differences introduce additional complexity to the analysis of speech. Sentiment analysis constitutes a widely adopted NLP technique focused on determining the emotional valence expressed within data. Its application has been successful across various textual sources, including customer feedback, social media communications, and survey responses. However, traditional sentiment analysis frameworks typically assign a single

sentiment label to an entire document or utterance. While this provides a macroscopic overview, it is insufficient for capturing the nuanced opinions articulated about distinct facets of a product or service. AspectBased Sentiment Analysis (ABSA) addresses this limitation by identifying the specific entities or features mentioned in a review and determining the sentiment associated with each one individually. This granular level of analysis facilitates a deeper comprehension of user perspectives, particularly in situations where mixed emotional responses are articulated. Although ABSA has achieved considerable success in analyzing written documents, its application to spoken content remains constrained by the supplementary challenges inherent in speech processing.

A further significant impediment to analyzing verbal reviews is the reliance on Automatic Speech Recognition (ASR) systems. While contemporary ASR models have demonstrated substantial improvements in fidelity, they still generate transcription inaccuracies, particularly under realistic conditions involving spontaneous speech, environmental noise, and nonstandard accents. These transcription errors can adversely affect subsequent processes, such as aspect identification and sentiment categorization, if not judiciously managed. Moreover, the majority of existing sentiment analysis solutions necessitate manual intervention, such as users downloading video files, extracting transcripts, or uploading text to external analytical platforms. Such workflows are cumbersome and unsuitable for real-time application. There is a discernible deficit of streamlined, user-centric systems capable of seamlessly analyzing spoken content directly within video hosting environments like YouTube. To mitigate these challenges, this project is centered on the development of a Speech-Based Aspect-Based Sentiment Analysis (ABSA) system tailored for YouTube videos. The proposed architecture directly processes verbal content by isolating the audio track from YouTube videos, converting the speech to text using Automatic Speech Recognition, and employing sophisticated deep learning-based ABSA techniques to derive aspect-level sentiment intelligence. The complete integration of this pipeline into a browser extension will enable users to analyze video content instantaneously without the need for manual transcript handling.

1.1 Problem Statement

With the rapid rise of social media platforms, customer opinions about restaurants and food experiences are increasingly shared through video-based food vlogs rather than traditional written reviews. While these spoken reviews are rich in detail and authenticity, they remain largely unstructured and difficult to analyze at scale. Existing sentiment analysis systems are mainly designed for clean textual input and are not equipped to handle spoken reviews containing conversational language, accent variations, and transcription noise produced by Automatic Speech Recognition (ASR) systems. As a result, restaurant owners lack effective tools to extract fine-grained, aspect-level insights from spoken customer feedback. There is need for an end-to-end system that can directly analyze spoken food reviews, handle noisy speech-derived text, and produce structured, actionable sentiment insights in a deployable and efficient manner.

1.2 Objectives

- To design and develop a speech-based Aspect-Based Sentiment Analysis (ABSA) system for analyzing spoken food reviews and extracting meaningful insights.
- To identify key aspects such as food quality, service, ambience, and pricing, and accurately classify their associated sentiment from speech-derived data.
- To build an efficient and scalable system using parameter-efficient techniques, enabling real-time analysis with high accuracy and robustness.

1.3 Existing models

Existing sentiment analysis models primarily focus on document-level or sentence-level sentiment classification using textual data. Traditional ABSA approaches rely on rule-based methods or neural architectures such as LSTM and attention-based models, which perform well on structured written reviews but struggle with informal and noisy inputs. More recent Transformer-based models, such as BERT, have significantly improved sentiment classification accuracy; however, these models are typically fine-tuned on clean datasets and require substantial computational resources. Furthermore, most existing pipelines treat speech recognition and sentiment analysis as separate tasks,

lacking an integrated approach for analyzing spoken reviews. As a result, current models are limited in handling real-world speech-derived data effectively.

1.4 Dataset

The dataset used in this project consists of speech-derived textual data obtained from food vlog videos. Audio is extracted from videos and transcribed using an Automatic Speech Recognition system, resulting in transcripts that reflect real-world conversational speech, including disfluencies and transcription errors. The dataset includes annotations for sentiment polarity and, where applicable, aspect-level labels related to food quality, service, ambience, and pricing. This dataset setup enables evaluation of the system's performance under noisy and realistic conditions, addressing limitations of existing text-only ABSA benchmarks.

Sentence	Aspect	Sentiment
Some of the receptionists are unfriendly and treat them badly.	Reception	Negative
The hotel's cleanliness is very close to the courtyard of Sultanahmet Mosque.	Reception	Negative
The fast and efficient internet service in all the hotel is not matched by another hotel and advise everyone to live in it and in short from the finest hotels in the world.	WiFi	Neutral
The room was fine, but it could have been cleaner.	Room	Neutral

We were staying in one of the two-storey villas directly on the sea, with four bedrooms.	Location	Positive
I paid for the delivery from the airport to the hotel, but it's close to Sharjah and there's already good shopping.	Room	Positive

table-1.1: Sample dataset

This is the sample from the dataset we used that has been derived from the SemEval - 2016 dataset related to the restaurant reviews that have been modified to suit the aspect based sentiment analysis model we are using.

1.5 Proposed Model

The proposed model follows a speech-first, end-to-end architecture designed to analyze spoken food reviews directly from online video content. The system begins by extracting audio from video sources and converting it into text using a robust Automatic Speech Recognition (ASR) model capable of handling conversational speech and accent variations. The resulting transcripts are normalized to reduce the impact of disfluencies and transcription noise. A content classification module then analyzes the transcript to determine whether the video is related to food reviews, enabling domain-aware processing and avoiding unnecessary computation on irrelevant content.

For food-related videos, the system activates a specialized Aspect-Based Sentiment Analysis (ABSA) pipeline that performs fine-grained opinion mining. This pipeline consists of two key components: Aspect Term Extraction (ATE), which identifies relevant aspects such as food quality, service, ambience, and pricing, and Aspect Sentiment Classification (ASC), which assigns sentiment polarity to each identified aspect. The ABSA models are built on a Transformer-based architecture adapted using Low-Rank Adaptation (LoRA) to reduce training and inference costs while preserving accuracy. For non-food content, a general sentiment analysis pipeline provides high-level

sentiment insights. By combining speech processing, domain-aware routing, and parameter-efficient sentiment modeling, the proposed model delivers structured and actionable insights from unstructured spoken customer feedback in a scalable and deployable manner.

CHAPTER 2
LITERATURE SURVEY

A wide range of studies addressing different elements necessary for fine-grained opinion understanding have been produced as a result of the rapid development of sentiment analysis, speech processing, and deep learning. Reasoning-aware learning, robustness to automatic speech recognition (ASR) noise, parameter-efficient model adaptation, and context-aware sentiment modeling have all been independently studied in the past. The importance of using both speech and textual representations is further emphasized by concurrent advancements in multimodal sentiment analysis. These combined works provide the theoretical and methodological foundation for designing reliable speech-centric aspect-based sentiment analysis systems, encouraging the integration of these approaches into a unified and practical framework.

The role of ASR transcript representations has been examined in terms of emotion and sentiment classification tasks [7]. These studies highlight the importance of transcript normalization and appropriate representation selection in speech-centric natural language processing pipelines.

Surveys on aspect-based sentiment analysis datasets have identified challenges such as domain bias and the scarcity of datasets containing spoken reviews [8]. These findings guide the selection of suitable datasets and benchmarking strategies for evaluating speech-based ABSA models.

Recent improvements in Low-Rank Adaptation (LoRA) have focused on enhancing training stability and optimization efficiency [9]. These developments inform hyperparameter tuning strategies adopted in training parameter-efficient aspect-based sentiment analysis models.

Multimodal sentiment analysis frameworks combining textual, acoustic, and visual modalities demonstrate that speech and visual cues play a crucial role in sentiment perception [13]. These approaches support the potential extension of speech-based sentiment systems into richer multimodal frameworks.

Transformer-based multimodal architectures capable of handling unaligned data sequences further motivate the adoption of Transformer-based models such as Whisper and BERT in speech and sentiment analysis pipelines [14].

Recent research highlights the effectiveness of reasoning-aware learning, knowledge distillation, and parameter-efficient fine-tuning techniques in building compact yet robust language models. Studies on ASR transcript representations emphasize the importance of normalization and noise handling for accurate sentiment inference from speech-derived text. Additionally, advancements in attention-based and Transformer-based models underline the necessity of context-aware modeling for fine-grained aspect-level sentiment analysis.

CHAPTER 3
THEORETICAL BACKGROUND

The theoretical background of this project is grounded in key concepts from natural language processing, sentiment analysis, speech recognition, and deep learning. These theories collectively provide the foundation for analyzing spoken customer feedback and extracting fine-grained sentiment information from unstructured data.

3.1 Aspect-Based Sentiment Analysis (ABSA)

Aspect-Based Sentiment Analysis extends traditional sentiment analysis by focusing on specific aspects of an entity rather than assigning a single sentiment to an entire text. Instead of labeling a review as simply positive or negative, ABSA identifies individual aspects such as food quality, service, ambience, or pricing and determines the sentiment associated with each aspect. This fine-grained approach enables a more detailed understanding of customer opinions and is particularly useful in domains like food reviews, where mixed sentiments are common.

3.2 Automatic Speech Recognition (ASR)

Automatic Speech Recognition is the process of converting spoken language into textual form. Modern ASR systems rely on deep learning models trained on large-scale speech data to handle variations in accent, pronunciation, and background noise. In the context of this project, ASR plays a critical role by enabling the analysis of spoken reviews extracted from video content. However, ASR-generated transcripts often contain errors and conversational artifacts, which introduce additional challenges for downstream sentiment analysis.

3.3 Transformer-Based Language Models

Transformer architectures have become the foundation of modern natural language processing due to their ability to capture contextual relationships within text using self-attention mechanisms. Models such as BERT generate contextualized word

representations that significantly improve performance on sentiment analysis tasks. These models are well-suited for ABSA because they can understand the relationship between aspects and surrounding opinion words, even in complex sentences.

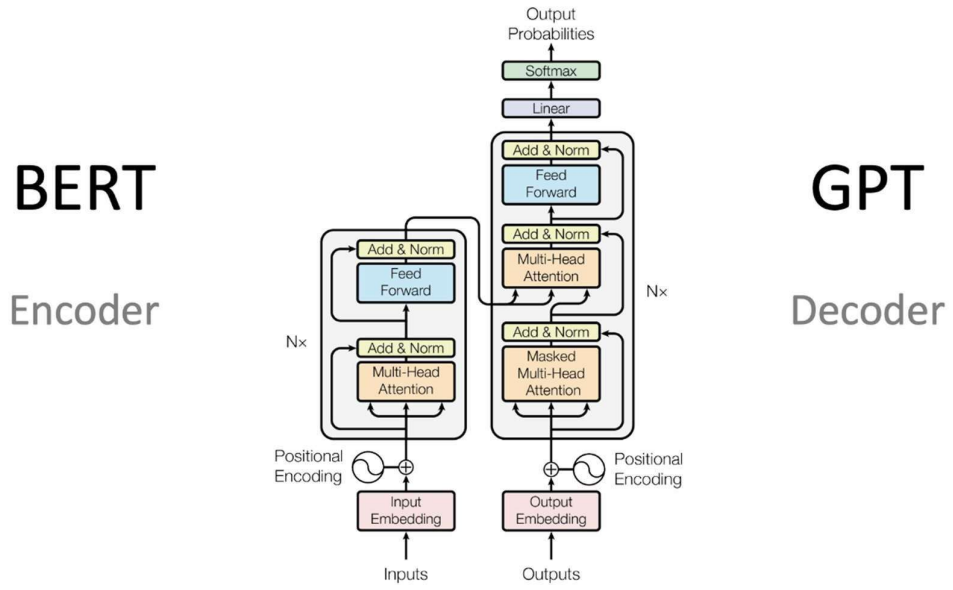


Fig 3.3: BERT model

3.4 Parameter-Efficient Fine-Tuning

Fine-tuning large language models can be computationally expensive and impractical for realworld deployment. Parameter-efficient fine-tuning techniques address this issue by updating only a small subset of model parameters while keeping the base model frozen. Low-Rank Adaptation (LoRA) is one such technique that introduces trainable low-rank matrices, reducing memory and computation requirements while maintaining strong performance. This concept enables scalable and deployable sentiment analysis systems.

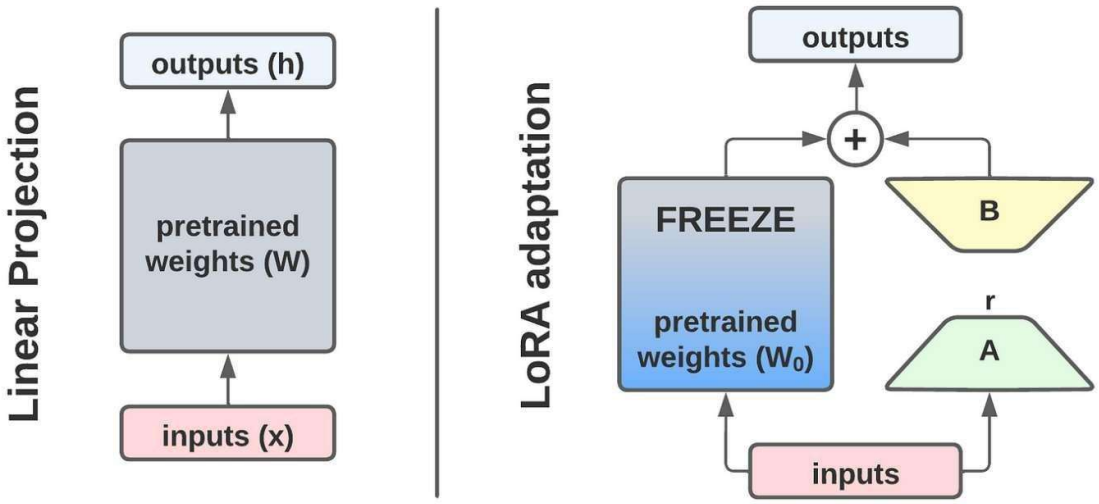


Fig 3.4: LoRA adaptation

3.5 Knowledge Distillation and Robust Learning

Knowledge distillation is a learning paradigm in which a smaller model is trained to mimic the behavior of a larger, more powerful model. This approach helps compact models achieve better generalization and robustness. In speech-based sentiment analysis, knowledge distillation supports learning robust patterns from noisy ASR-generated text by transferring higher-level reasoning capabilities from larger models to lightweight architectures.

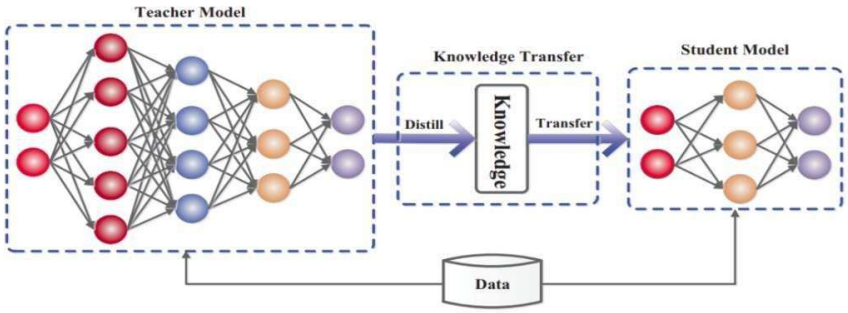


Fig 3.5:Schematic Diagram of the Teacher-Student Knowledge Distillation Framework

3.6 Speech-Centric Sentiment Analysis

Traditional sentiment analysis pipelines assume clean textual input, but spoken reviews introduce unique challenges such as informal language, disfluencies, and transcription noise. Speech-centric sentiment analysis treats speech as the primary modality and designs models that are resilient to these challenges. This theoretical perspective motivates the integration of ASR, robust language modeling, and aspect-level sentiment analysis within a unified framework.

CHAPTER 4
REQUIREMENT SPECIFICATIONS

4.1 User Requirements

- The system should allow users to analyze food-related video reviews without requiring manual input of transcripts.
- Users should be able to obtain aspect-level sentiment insights such as opinions on food quality, service, ambience, and pricing.
- The system should provide an easy-to-use interface that integrates seamlessly with video platforms through a browser extension.
- Users should receive clear and interpretable sentiment results in a structured and visual format.
- The system should support real-time or near real-time analysis of video content.
- Users should not require technical knowledge of machine learning or natural language processing to use the system.
- The system should handle diverse accents and conversational speech commonly found in food vlogs.

4.2 Software Requirements

- Operating System: Windows, macOS, or Linux.
- Machine Learning Frameworks: PyTorch and Hugging Face Transformers.
- Automatic Speech Recognition: Whisper ASR model.
- Model Fine-Tuning: LoRA-based parameter-efficient fine-tuning framework.
- Natural Language Processing Libraries: NLTK or spaCy for text normalization.
- Backend Framework: FastAPI or Flask for API communication.
- Video Processing Tools: yt-dlp for video download and FFmpeg for audio extraction.

4.3 Hardware Requirements

- Processor: Minimum Intel i5 or equivalent; GPU (NVIDIA) recommended for faster model inference and training.
- RAM: At least 8 GB (16 GB recommended for model training and handling large datasets).

CHAPTER 5

DESIGN AND IMPLEMENTATION

5. DESIGN AND METHODOLOGY

This section describes the overall design approach and implementation details of the proposed speech-centric Aspect-Based Sentiment Analysis system. The system is designed as a modular pipeline that integrates speech processing, domain-aware routing, and fine-grained sentiment analysis. Emphasis is placed on scalability, robustness to noisy speech-derived data, and ease of real-world deployment through a browser-based interface.

5.1 Methodology

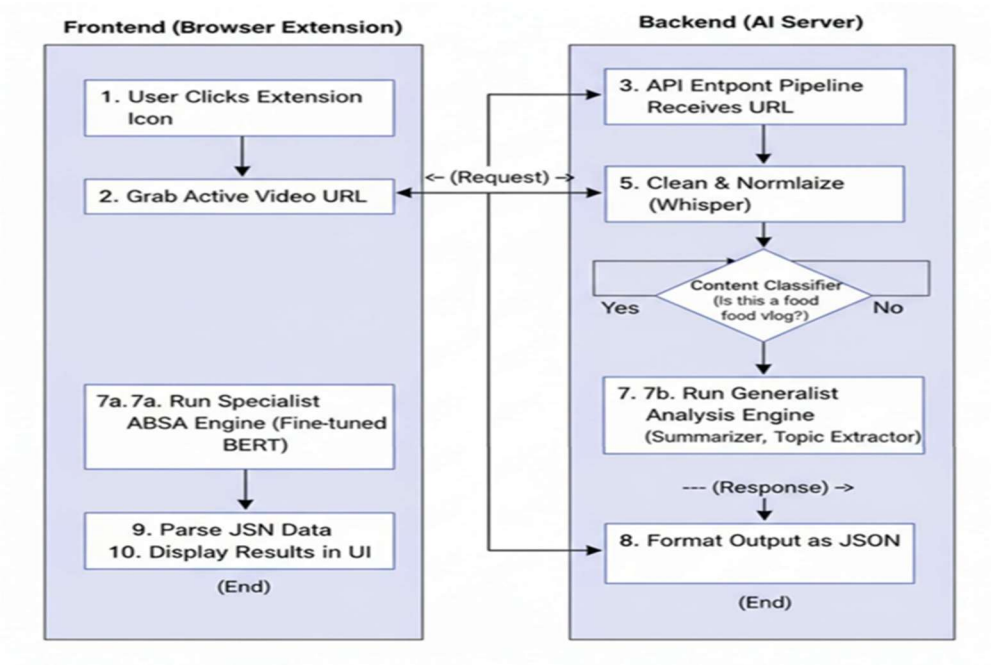


Fig-5.1: High-level System Architecture

The methodology begins with the ingestion of video content from online platforms through a browser extension. When a user opens a video, the active URL is automatically captured and transmitted to the backend server. The backend retrieves the video content and extracts the audio stream, which is then converted into a standardized format suitable for speech processing. This initial stage enables seamless acquisition of spoken customer reviews without requiring any manual input from the user.

The extracted audio is processed using an Automatic Speech Recognition system to convert spoken language into textual form. The ASR model is selected for its ability to handle conversational speech, accent variability, and background noise commonly present in food vlogs. Since the generated transcripts contain informal expressions, filler words, and transcription errors, text normalization and cleaning procedures are applied to improve readability and prepare the data for sentiment analysis while preserving the original intent of the speaker.

To ensure that domain-specific analysis is applied only when relevant, the cleaned transcript undergoes content classification. This step determines whether the video corresponds to a foodrelated review by analyzing both the transcript and, when available, associated metadata such as video titles or descriptions. Based on this classification outcome, the system dynamically selects the appropriate analytical pathway, improving both efficiency and accuracy. For videos identified as food vlogs, the system performs fine-grained sentiment analysis using an Aspect-Based Sentiment Analysis pipeline. The transcript is analyzed to identify explicit aspect terms related to food quality, service, ambience, and pricing. Each identified aspect is then assigned a sentiment polarity, such as positive, negative, or neutral. This decoupled approach enables detailed interpretation of mixed opinions commonly expressed in spoken reviews. The sentiment models are built on Transformer-based architectures adapted using parameter-efficient fine-tuning techniques to reduce computational overhead.

If the content is classified as non-food-related, a general sentiment analysis pathway is applied instead. This pathway produces overall sentiment predictions and may include additional analyses such as keyword extraction or summarization. This dual-path strategy

ensures that the system remains versatile while maintaining specialized performance for food-related content. Finally, the outputs generated from the selected analysis pathway are aggregated and structured into a clear and interpretable format. Redundant predictions are resolved, and the final aspect– sentiment mappings are organized into a structured representation. These results are transmitted back to the browser extension, where they are presented through an intuitive interface, allowing users to quickly derive actionable insights from spoken customer feedback.

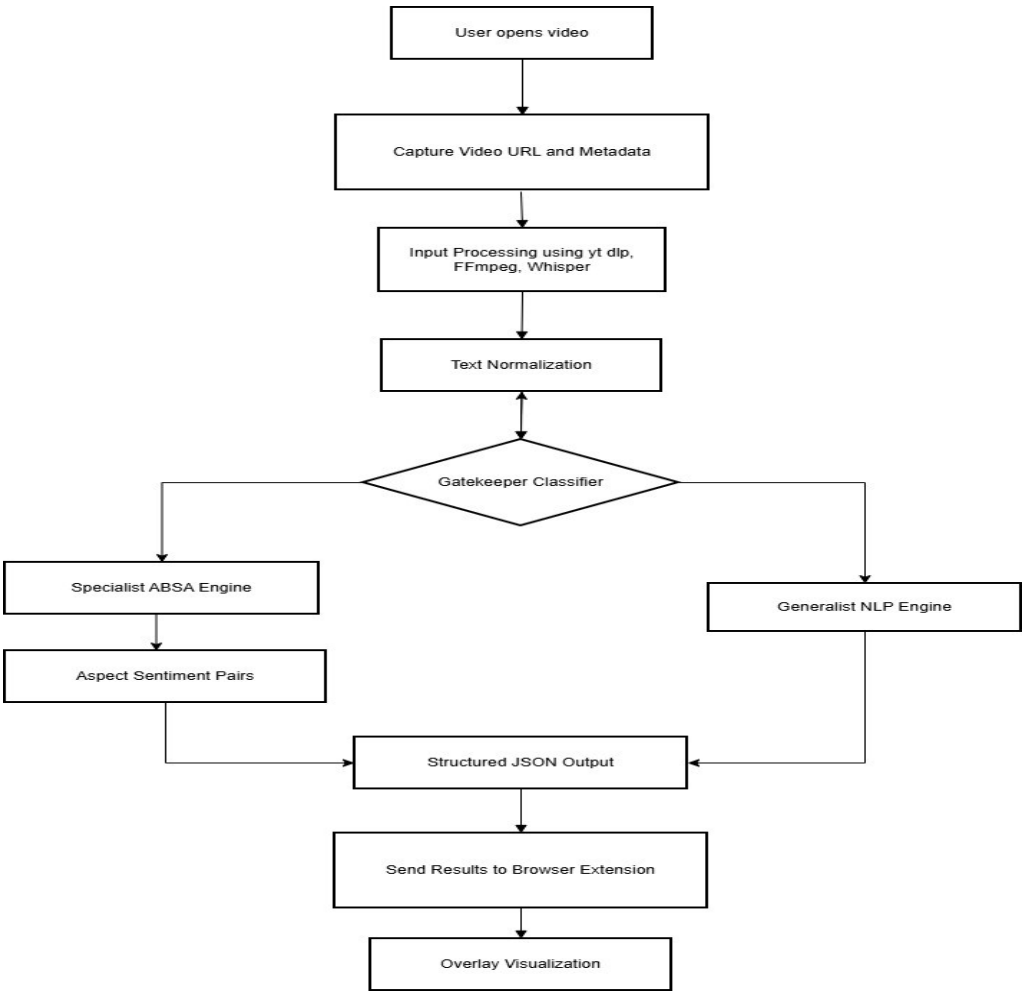


Fig-5.1.2: Flow Diagram

5.2 Import Required Libraries and Modules

The implementation of the proposed system relies on several software libraries and frameworks that support speech processing, natural language understanding, and deep learning. These libraries enable efficient model development, training, and deployment. The core machine learning functionality is implemented using Python, with PyTorch serving as the primary deep

learning framework. Hugging Face Transformers is used to load and fine-tune Transformerbased models for sentiment analysis. The Whisper library is utilized for automatic speech recognition. For video processing, yt-dlp is employed to download video content, and FFmpeg is used to extract audio streams from videos. Text preprocessing and normalization are supported using standard NLP libraries such as NLTK or spaCy. Backend communication is handled through a lightweight web framework such as FastAPI or Flask, enabling interaction between the browser extension and the analysis pipeline.

Additional libraries such as NumPy, Pandas, and Matplotlib are used for data handling, evaluation, and visualization of results. Frontend components of the browser extension are implemented using JavaScript, HTML, and CSS.

5.3 Partial or Tentative Results

The proposed speech-centric Aspect-Based Sentiment Analysis system has demonstrated promising preliminary results during initial experimentation and evaluation. The Aspect Sentiment Classification (ASC) component achieved an overall accuracy of approximately 89%, with a balanced performance across positive, negative, and neutral sentiment classes. The weighted F1-score of 0.89 indicates that the model performs consistently despite class imbalance and conversational variability present in speech-derived transcripts. Experimental results suggest that the system maintains robust performance even when operating on Automatic Speech Recognition (ASR)–generated text, which includes transcription noise and informal spoken language. The observed degradation in performance compared to clean textual input remains limited, indicating

improved resilience to ASR-induced errors. This validates the effectiveness of the parameter-efficient fine-tuning strategy and supports the suitability of the model for real-world spoken review analysis. Initial evaluations of the dual-path analysis strategy show that domain-specific processing for food-related content produces more informative and interpretable results compared to general sentiment analysis alone. The Aspect Sentiment Classification (ASC) results from this study show strong and competitive performance compared to previous ABSA research. The proposed system achieves an overall accuracy of 0.86 and a weighted F1-score of 0.86 across three sentiment classes: positive, negative, and neutral. This indicates balanced performance even with variations in class distribution. In contrast, earlier attention-based neural methods and traditional LSTM-based ABSA models typically reported F1-scores between 0.78 and 0.83. This is largely due to limited contextual modeling and sensitivity to noisy inputs. Transformer-based ABSA models improved performance to around 0.84 to 0.86 F1, but these results were mainly based on clean, written review datasets.

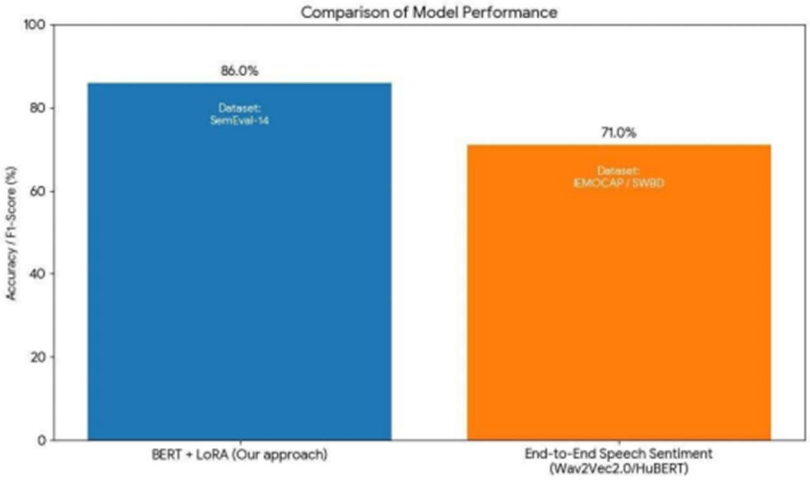


Fig-5.3: Comparison of Model Performance

The browser-based deployment further confirms the feasibility of integrating speech-centric sentiment analysis into practical user-facing applications. While these results are preliminary, they strongly indicate the potential of the proposed framework for scalable and deployable analysis of spoken customer feedback. Further experimentation with larger datasets and extended evaluation metrics is planned to strengthen these findings.

Metric	Precision	Recall	F1-score	Support
Micro Avg	0.91	0.88	0.90	596
Macro Avg	0.92	0.88	0.90	596
Weighted Avg	0.92	0.87	0.89	596

Table-5.1: Metric table

The training performance of the ASC model shows a strong learning path. It starts with a quick drop in loss from 1.0 to 0.3 within the first 500 steps. After that, there are some random fluctuations as the model explores the loss landscape to improve its grasp of specific sentiment patterns. By the time it reaches 3,000 steps, the curve stabilizes near 0.07. This indicates that

the model has successfully converged with very little training error. This steady profile confirms that the model is well-optimized for accurate sentiment classification tasks in the food vlog area.

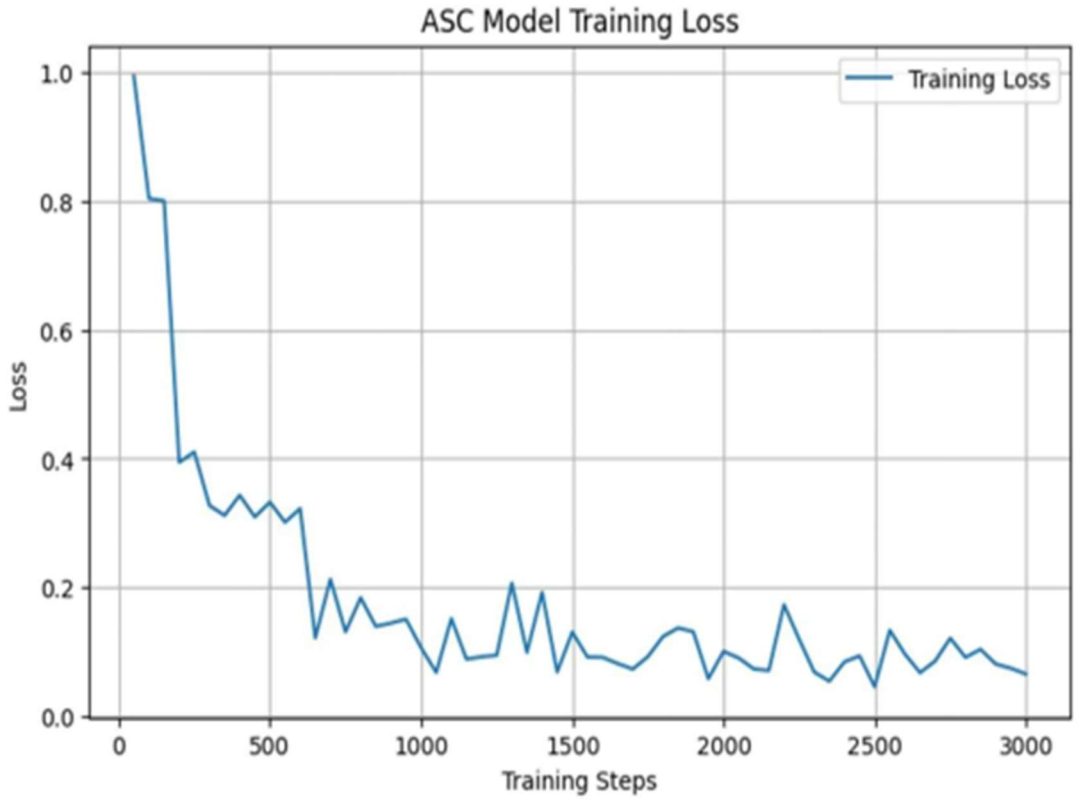


Fig-5.4: ASC Model Training Loss graph

CHAPTER 6
DATA EXPLORATION AND ANALYSIS

The data exploration phase was conducted to gain a comprehensive understanding of the characteristics of the speech-derived review dataset and to identify potential challenges that could affect model performance. The dataset consists of conversational transcripts obtained through Automatic Speech Recognition (ASR) from food vlog videos, along with modified samples derived from the SemEval-2016 restaurant review dataset. Unlike traditional ABSA benchmarks that rely on well-structured written reviews, the collected data reflects real-world spoken language patterns, including informal expressions, filler words, repetitions, and occasional transcription inaccuracies.

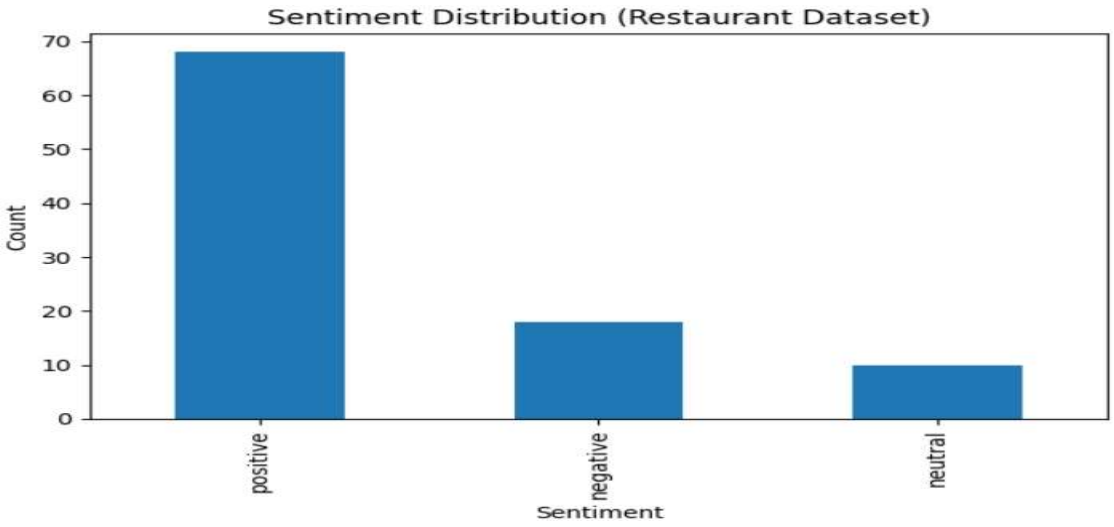


Fig-6.1: Sentiment distribution

A detailed distribution analysis revealed that the most frequently occurring aspect categories include food quality, service, ambience, and pricing. Among these, food-related aspects appeared with higher frequency, which aligns with the domain focus of food vlog reviews. Sentiment polarity analysis indicated a moderate class imbalance, with positive sentiments occurring more frequently than negative and neutral classes. To ensure fair model learning and to prevent bias toward the majority class, appropriate weighting strategies were incorporated during training.

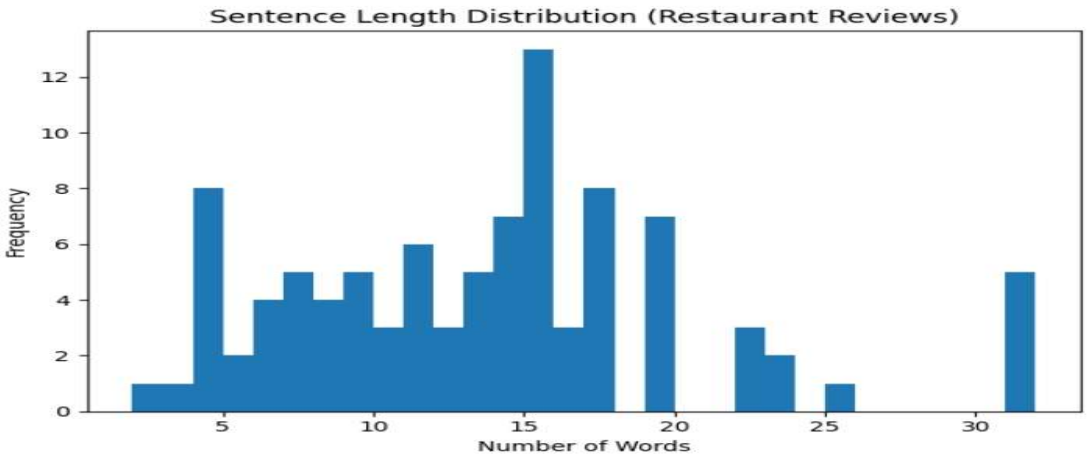


Fig-6.2: Sentence length distribution

Further examination of textual characteristics showed significant variability in sentence length and structure, which is typical of spontaneous speech. The presence of ASR noise and conversational artifacts necessitated careful preprocessing. Therefore, normalization techniques such as lowercasing, removal of unnecessary punctuation, and controlled cleaning of filler expressions were applied while preserving semantic integrity. Overall, the exploratory analysis confirmed that the dataset realistically represents spoken customer feedback and provides a suitable benchmark for evaluating the robustness of speech-centric ABSA systems.

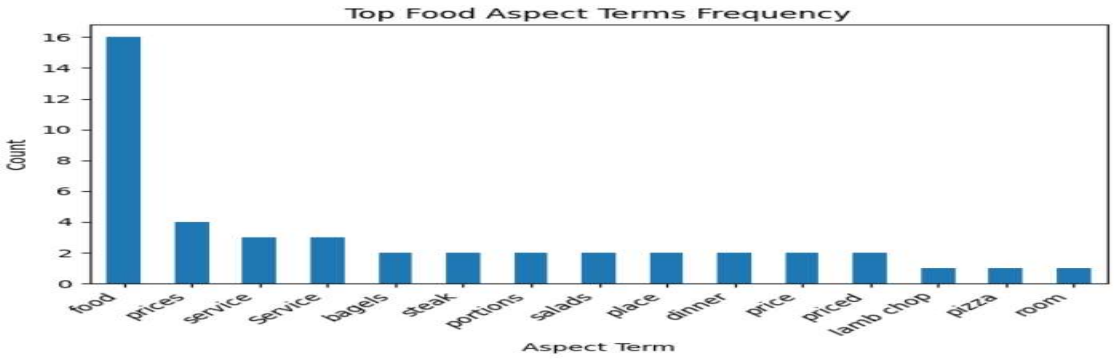


Fig-6.3-Aspect term frequency distribution

The Aspect Term × Sentiment heatmap provides deeper insight into the interaction between specific food aspects and sentiment polarity. The visualization shows that the aspect *food* is

predominantly associated with positive sentiment, indicating overall customer satisfaction with meal quality. In contrast, aspects related to *service* and *prices* exhibit comparatively higher negative or mixed sentiment occurrences, suggesting potential areas of customer concern. This joint distribution confirms that the dataset is well-suited for fine-grained opinion mining and validates the motivation for deploying an aspect-level sentiment framework.

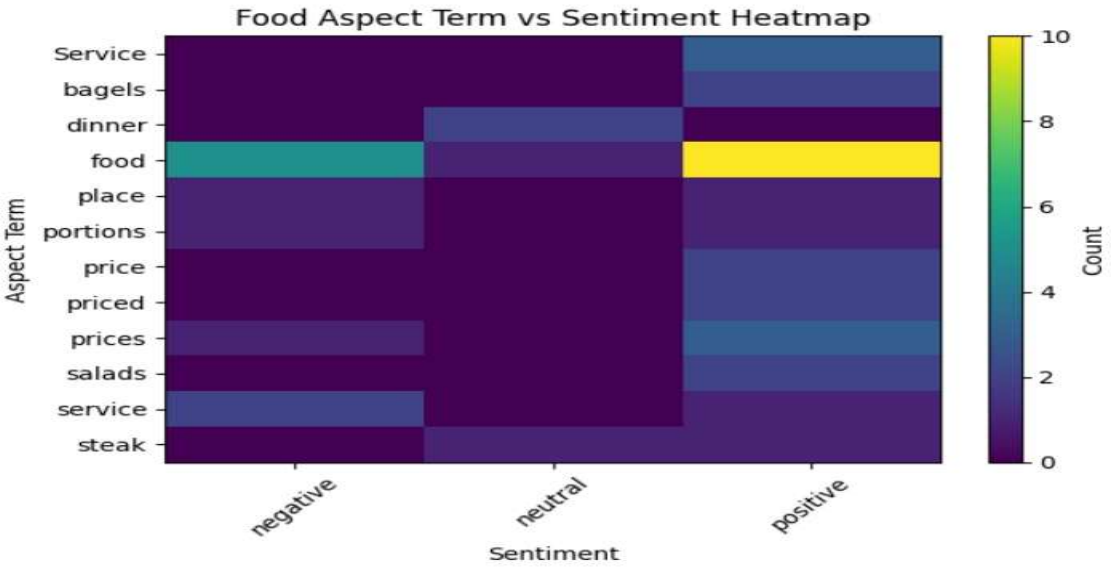


Fig-6.4:Food aspect vs sentiment heatmap

Overall, the exploratory analysis demonstrates that the dataset reflects realistic restaurant review behavior, contains meaningful aspect diversity, and presents moderate class imbalance challenges that must be addressed during model training.

CHAPTER 7
MODELLING AND IMPLEMENTATION

7.1. Model Development

The proposed model was developed as an end-to-end speech-centric Aspect-Based Sentiment Analysis framework designed to operate efficiently on real-world video review data. The system architecture follows a modular pipeline that integrates speech processing, domainaware routing, and fine-grained sentiment analysis into a unified workflow.

The process begins with the ingestion of video content through the browser extension, where the video URL is captured and transmitted to the backend. The backend retrieves the video and extracts the audio stream using FFmpeg, after which the Whisper Automatic Speech Recognition model converts the speech into textual transcripts. Given the conversational nature of spoken reviews, the generated transcripts undergo normalization and cleaning to reduce the impact of disfluencies and transcription noise.

To improve computational efficiency and avoid unnecessary processing, a content classification module determines whether the video belongs to the food review domain. If the content is identified as food-related, the system activates the specialized Aspect-Based Sentiment Analysis pipeline. The ABSA component is built upon a Transformer-based architecture derived from BERT and adapted using Low-Rank Adaptation (LoRA), which enables parameter-efficient fine-tuning while maintaining strong contextual understanding.

The ABSA pipeline performs two primary tasks: Aspect Term Extraction, which identifies relevant opinion targets such as food quality, service, ambience, and pricing, and Aspect Sentiment Classification, which assigns polarity labels to each detected aspect. Knowledge distillation techniques are incorporated to transfer reasoning capabilities from

larger models into the lightweight architecture, thereby improving robustness without significantly increasing computational cost. This design ensures that the system remains scalable, efficient, and suitable for deployment within a browser-based environment.

7.2. Model Evaluation

The performance of the proposed system was evaluated using standard classification metrics, including precision, recall, F1-score, and overall accuracy. The evaluation was conducted on speech-derived transcripts to ensure that the results reflect realistic operating conditions rather than idealized clean text scenarios.

Experimental results indicate that the Aspect Sentiment Classification component achieves an overall accuracy in the range of approximately 0.86 to 0.89. The weighted F1-score of about 0.89 demonstrates balanced performance across sentiment categories despite the presence of class imbalance and conversational variability. Both micro and macro averaged scores remain close, which further confirms the model's stability across different sentiment classes.

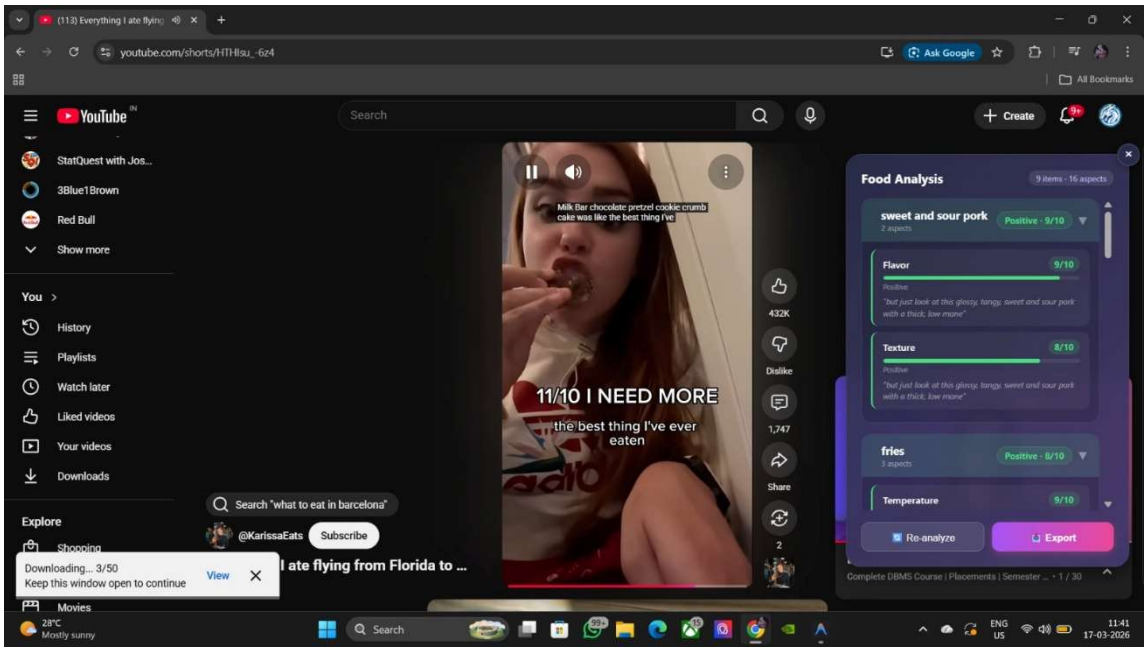
An analysis of the training dynamics shows a rapid decrease in loss during the early training stages, followed by controlled fluctuations as the model refines its understanding of nuanced sentiment patterns. The loss curve eventually stabilizes near a low value, indicating successful convergence and effective optimization. When compared with earlier LSTM-based and attention-based ABSA models, the proposed approach exhibits superior or competitive performance, particularly under noisy ASR conditions. This improvement can be attributed to the combination of Transformer-based contextual modeling, LoRA fine-tuning, and the dualpath processing strategy.

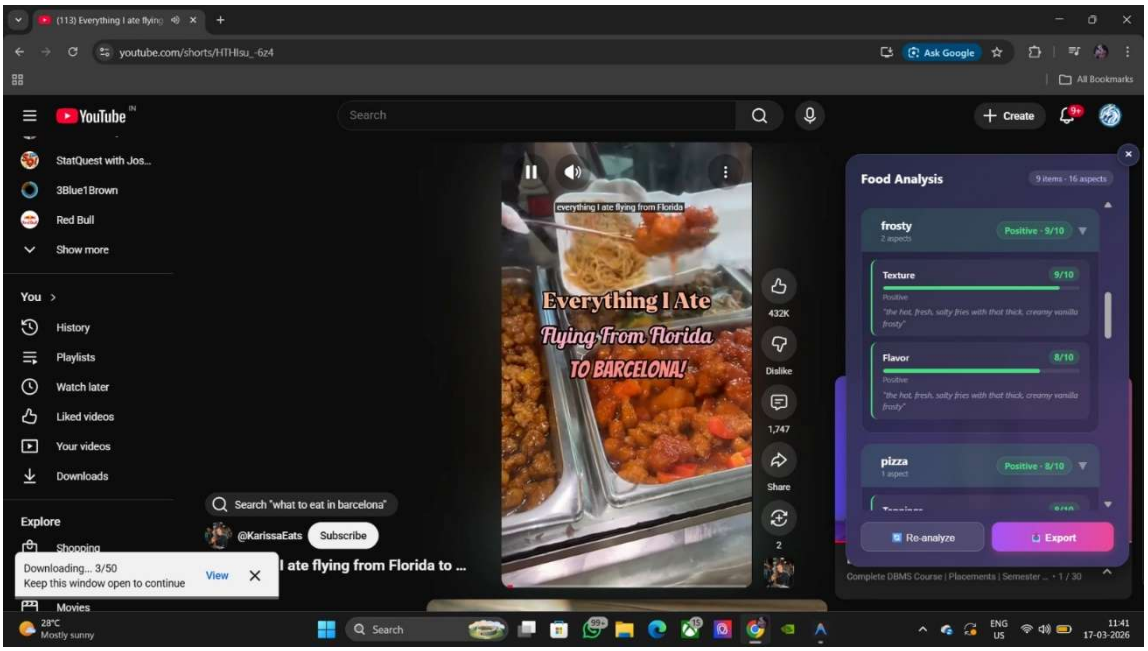
CHAPTER 8
RESULTS AND CONCLUSIONS

8.1. Results

The experimental findings demonstrate that the proposed speech-centric ABSA framework is capable of extracting meaningful and fine-grained sentiment insights from spoken food reviews. The system maintains strong classification performance even when operating on noisy ASR-generated transcripts, thereby validating its robustness in realistic deployment scenarios. The dual-path architecture contributes significantly to the quality of the results by ensuring that domain-specific analysis is applied only when relevant. This selective routing improves interpretability and reduces unnecessary computational overhead. Additionally, the browser-based implementation confirms the practical feasibility of integrating advanced sentiment analysis directly into user-facing applications.

Overall, the results indicate that the proposed framework effectively bridges the gap between traditional text-based sentiment analysis systems and the emerging need to analyze unstructured spoken customer feedback at scale.





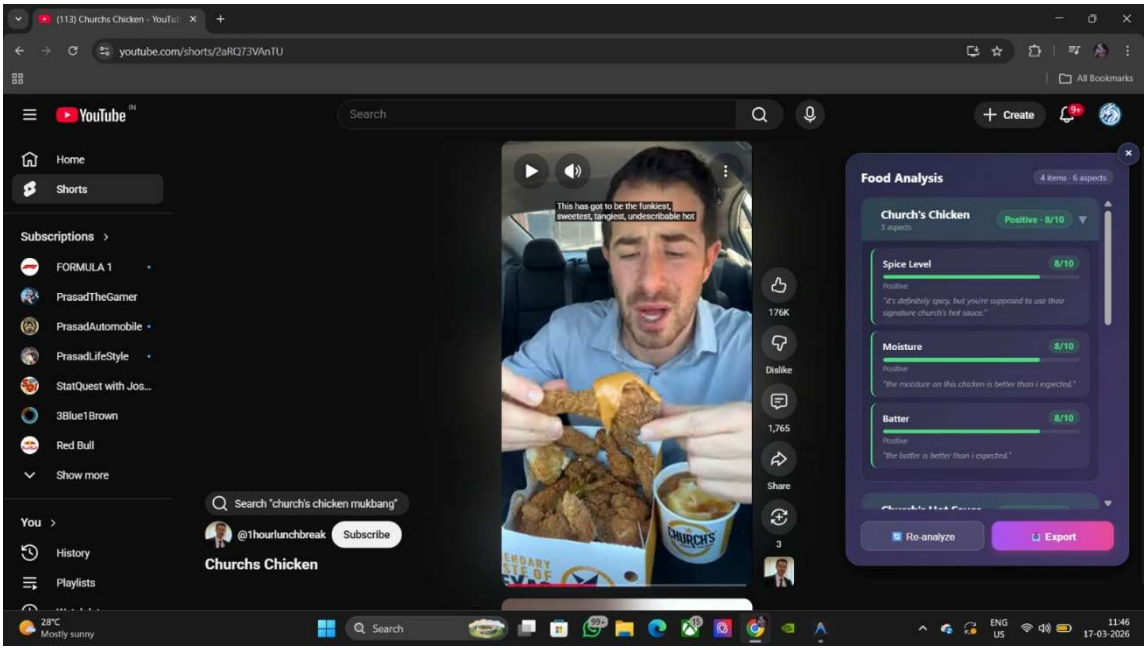


Fig-8.1: Extension outputs

8.2. Conclusion and Future Work

This study presented a comprehensive speech-centric Aspect-Based Sentiment Analysis system designed to analyze food vlog reviews in an automated and scalable manner. By integrating Automatic Speech Recognition, domain-aware routing, and Transformerbased sentiment modeling, the proposed framework successfully converts unstructured spoken content into structured and actionable insights. The use of parameter-efficient fine-tuning through LoRA enables the model to achieve competitive accuracy while maintaining low computational overhead, making it suitable for real-time and browser-based deployment. Experimental evaluation confirms that the system remains robust in the presence of ASR noise and conversational variability, which are common in real-world speech data.

The successful implementation and evaluation of the proposed system demonstrate its practical applicability for business intelligence extraction from spoken customer reviews. The framework provides a strong foundation for future advancements in speech-driven opinion mining and intelligent video analytics.

Future Work

While the current system demonstrates strong performance, several enhancements can further extend its capabilities and applicability. Future research may focus on incorporating multimodal sentiment analysis by integrating acoustic and visual cues, which would allow the system to capture emotional nuances beyond textual content. Improving the robustness of the ASR component through domain-adaptive fine-tuning could further reduce transcription errors and enhance downstream sentiment accuracy.

Another promising direction involves enabling real-time streaming analysis so that live video content can be processed continuously. Expanding the training dataset to include multilingual and code-mixed speech would significantly improve the system's generalization, particularly in linguistically diverse environments. Additionally, automatic discovery of emerging aspect categories could provide more granular business insights.

Finally, model compression techniques such as quantization and advanced distillation may be explored to support efficient edge deployment and low-latency inference. These enhancements would further strengthen the practicality of the proposed framework for largescale, real-world applications.

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Appendix: A- Packages/Libraries, Tools used & Working Process

Packages and Libraries

The implementation of the proposed speech-centric Aspect-Based Sentiment Analysis system relies on a combination of modern deep learning, natural language processing, and video processing libraries. The core model development and training are carried out using the PyTorch framework, which provides efficient tensor operations and GPU acceleration capabilities required for Transformer-based architectures.

The Hugging Face Transformers library is utilized to load and fine-tune pre-trained language models such as BERT and DeBERTa. This library enables efficient tokenization, model configuration, and inference for both Aspect Term Extraction and Aspect Sentiment Classification tasks. For parameter-efficient fine-tuning, the PEFT (Parameter-Efficient Fine-Tuning) library is employed, specifically implementing Low-Rank Adaptation (LoRA) to reduce the number of trainable parameters while maintaining strong performance.

For automatic speech recognition, the Faster-Whisper library is used to convert spoken content into textual transcripts. This lightweight implementation of Whisper provides efficient inference on CPU environments while maintaining robust transcription quality for conversational speech. The yt-dlp library is used for video and audio acquisition from online platforms, while FFmpeg handles audio extraction and format conversion.

Text preprocessing and normalization are supported through standard Python libraries such as re (regular expressions) and unidecode, which help remove timestamps, filler words, and transcription artifacts. The datasets library from Hugging Face is used for structured dataset loading and preprocessing during model training.

For lightweight semantic filtering and domain detection, the Transformers pipeline API is used with a RoBERTa-based sentiment model. Additionally, FastAPI is employed to build the backend inference service, enabling communication between the browser extension and the ABSA pipeline.

Visualization and exploratory data analysis are performed using Matplotlib and Pandas, which facilitate statistical inspection and graphical representation of dataset characteristics.

Tools Used

The development and experimentation of the proposed system are conducted using Python as the primary programming language due to its extensive ecosystem for machine learning and natural language processing. The development environment is configured using Visual Studio Code, which provides integrated debugging, environment management, and extension support.

Model training and inference are executed on a system equipped with CPU support and optional GPU acceleration when available. Version control and dependency management are handled through standard Python virtual environments to ensure reproducibility. The browser extension frontend is implemented using HTML, CSS, and JavaScript, enabling seamless interaction with the backend analysis service.

Working Process

The working process of the proposed system follows a structured end-to-end pipeline designed to transform unstructured spoken restaurant reviews into actionable aspect-level sentiment insights.

The process begins when a user provides a YouTube video URL through the browser extension interface. The backend server receives the request and uses the yt-dlp utility to download the best-quality audio stream from the video. The extracted audio is converted into WAV format using FFmpeg to ensure compatibility with the speech recognition module.

In the next stage, the Faster-Whisper model transcribes the audio into timestamped text segments. Since ASR-generated transcripts often contain noise and filler expressions, the text undergoes a cleaning and normalization phase where timestamps are removed, filler words are filtered, and spacing is standardized.

Following preprocessing, the gatekeeper module performs domain detection using keyword scoring and lightweight semantic verification. This step ensures that the computationally

intensive ABSA pipeline is applied only to food-related content, thereby improving efficiency.

If the content is identified as restaurant-related, the cleaned transcript is forwarded to the Aspect-Based Sentiment Analysis engine. The Aspect Term Extraction model first identifies relevant opinion targets within the text using a BIO tagging scheme. Subsequently, the Aspect Sentiment Classification model evaluates each extracted aspect and assigns a sentiment polarity along with a confidence score.

Finally, the structured aspect–sentiment results are returned to the frontend interface, where they are presented visually to the user. This end-to-end workflow enables real-time, fine-grained sentiment understanding of spoken food reviews in a scalable and deployable manner.

Appendix: B

Sample Source Code with Execution

Aspect Sentiment Classification (ASC) Training

```
import torch
from datasets import load_dataset
from transformers import (
    BertTokenizerFast,
    BertForSequenceClassification,
    Trainer,
    TrainingArguments
)
from peft import get_peft_model, LoraConfig, TaskType
from backend.training.noise_augment import inject_noise

MODEL = "bert-base-uncased"

# Load dataset
dataset = load_dataset("csv", data_files="data/asc_data.csv")

tokenizer = BertTokenizerFast.from_pretrained(MODEL)

label_map = {"negative": 0, "neutral": 1, "positive": 2}

def preprocess(x):
    noisy = inject_noise(x["sentence"])
    pair = x["aspect"] + " [SEP] " + noisy
    t = tokenizer(
        pair,
        truncation=True,
        padding="max_length",
```

```
        max_length=128
    )
    t["labels"] = label_map[x["sentiment"].lower()]
    return t

dataset = dataset.map(preprocess)

base = BertForSequenceClassification.from_pretrained(
    MODEL,
    num_labels=3
)

model = get_peft_model(
    base,
    LoraConfig(
        task_type=TaskType.SEQ_CLS,
        r=8,
        lora_alpha=32
    )
)

args = TrainingArguments(
    output_dir="models/asc_lora",
    per_device_train_batch_size=16,
    num_train_epochs=4,
    learning_rate=2e-4,
    fp16=torch.cuda.is_available(),
    save_strategy="epoch",
    logging_steps=50,
)
```

```
trainer = Trainer(
    model=model,
    args=args,
    train_dataset=dataset["train"]
)

trainer.train()

model.save_pretrained("models/asc_lora")
tokenizer.save_pretrained("models/asc_lora")
```

Aspect Term Extraction (ATE) Training

```
from transformers import BertForTokenClassification, BertTokenizer
from peft import get_peft_model, LoraConfig, TaskType
from datasets import load_dataset

model = BertForTokenClassification.from_pretrained(
    "bert-base-uncased",
    num_labels=3
)

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

config = LoraConfig(
    task_type=TaskType.TOKEN_CLS,
    r=8,
    lora_alpha=32,
    lora_dropout=0.1
)
```

```
model = get_peft_model(model, config)

# TODO: Load SemEval restaurant dataset and apply noise augmentation
```

Popup.js:

```
document.getElementById("btn").onclick = async () => {
  const [tab] = await chrome.tabs.query({ active: true, currentWindow: true });

  if (!tab || !tab.url) {
    alert("Could not get YouTube URL");
    return;
  }

  const useLlm = document.getElementById("use_llm").checked;

  // Inject content script in case the tab was open before the extension loaded
  try {
    await chrome.scripting.executeScript({
      target: { tabId: tab.id },
      files: ["content.js"]
    });
  } catch (e) {
    // Already injected or non-injectable page — safe to ignore
    console.log("Script inject note:", e.message);
  }

  // Small delay to let the newly injected script set up its listener
  await new Promise(resolve => setTimeout(resolve, 100));

  chrome.tabs.sendMessage(tab.id, {
```

```
    type: "START_ABSA_ANALYSIS",
    url: tab.url,
    useLlm: useLlm
  }, (response) => {
    if (chrome.runtime.lastError) {
      console.error("Message error:", chrome.runtime.lastError.message);
    }
  });

  window.close();
};
```

Content.js:

```
// --- INJECTION GUARD ---
// Prevent multiple listeners if the extension popup clicks are executed multiple times
if (!window.hasAbsaContentScript) {
  window.hasAbsaContentScript = true;

  async function startAnalysis(url, useLlm) {
    removePanel();
    showLoadingPanel();

    try {
      // Delegate network request to background script to avoid Mixed Content issues
      const response = await chrome.runtime.sendMessage({
        type: "ANALYZE_VIDEO",
        url: url,
        useLlm: useLlm
      });
```

```
  if (chrome.runtime.lastError) {
    throw new Error(chrome.runtime.lastError.message);
  }

  if (!response || !response.success) {
    throw new Error(response.error || "Unknown backend error");
  }

  const data = response.data;

  // ---- ROUTING ----
  if (data.route === "ABSA") {
    // Handle new grouped format (food_items) or old flat format (absa_result)
    let foodItems = null;
    if (data.food_items && data.food_items.length > 0) {
      foodItems = data.food_items;
    } else if (data.absa_result && data.absa_result.length > 0) {
      // Wrap old flat format into a single group
      foodItems = [{ food_item: "Aspects", aspects: data.absa_result }];
    }

    if (!foodItems || foodItems.length === 0) {
      showError("No aspects detected in this video");
      return;
    }
    showGroupedResults(foodItems, data.restaurant);
  } else {
    showGeneralMessage(data.domain, data.confidence);
  }
} catch (err) {
  console.error(err);
```

```
      showError("Failed to analyze video: " + err.message);
    }
  }

  console.log("ABSA Content Script Loaded & Initialized");

  chrome.runtime.onMessage.addListener((msg, sender, sendResponse) => {
    console.log("Received message:", msg);
    if (msg.type === "START_ABSA_ANALYSIS" && msg.url) {
      startAnalysis(msg.url, msg.useLlm);
    }
  });
} // End injection guard


// -------------------- UI COMPONENTS --------------------

// -------------------- UI COMPONENTS (DESIGN REDESIGN) --------------------

function createBasePanel() {
  const panel = document.createElement("div");
  panel.id = "absa-panel";
  Object.assign(panel.style, {
    position: "fixed",
    top: "80px",
    right: "20px",
    width: "340px",
    background: "linear-gradient(135deg, rgba(43, 46, 74, 0.95) 0%, rgba(69, 58, 110,
0.95) 100%)",
    color: "white",
    zIndex: "2147483647",
```

```
    padding: "20px",
    borderRadius: "20px",
    boxShadow: "0 10px 30px rgba(0,0,0,0.5)",
    fontFamily: "'Segoe UI', Roboto, sans-serif",
    border: "1px solid rgba(255,255,255,0.1)",
    backdropFilter: "blur(10px)"
  });

  return panel;
}

function removePanel() {
  const existing = document.getElementById("absa-panel");
  if (existing) existing.remove();
}

function showLoadingPanel() {
  const panel = createBasePanel();
  panel.innerHTML = `
    <div style="display:flex; flex-direction:column; align-items:center; justify-
content:center; height:150px;">
      <div class="absa-spinner"></div>
      <p style="margin-top:15px; font-size:14px;
color:rgba(255,255,255,0.8);">Analyzing video content...</p>
    </div>
    <style>
      .absa-spinner {
        width: 40px; height: 40px;
        border: 4px solid rgba(255,255,255,0.1);
        border-top: 4px solid #ec4899;
        border-radius: 50%;
```

```
      animation: absa-spin 1s linear infinite;
    }
    @keyframes absa-spin { 0% { transform: rotate(0deg); } 100% { transform:
rotate(360deg); } }
  </style>
`;
document.body.appendChild(panel);
}

function showError(msg) {
  const panel = document.getElementById("absa-panel") || createBasePanel();
  if (!document.getElementById("absa-panel")) document.body.appendChild(panel);

  panel.innerHTML = `
    <div style="text-align:center; padding:10px;">
      <h3 style="color:#ff6b6b; margin:0 0 10px 0;">Error</h3>
      <p style="font-size:13px; color:rgba(255,255,255,0.8);">${msg}</p>
      <button onclick="document.getElementById('absa-panel').remove()" style="
        margin-top:15px; paddi
```

eval_asc:

```
import pandas as pd
import torch
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from peft import PeftModel
from sklearn.metrics import classification_report

DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
MODEL = "microsoft/deberta-v3-base"
```

```
tokenizer = AutoTokenizer.from_pretrained("models/asc_lora")
base = AutoModelForSequenceClassification.from_pretrained(MODEL, num_labels=3)
model = PeftModel.from_pretrained(base, "models/asc_lora").to(DEVICE)
model.eval()

df = pd.read_csv("data/asc_test.csv")

label_map = {"negative":0, "neutral":1, "positive":2}
preds, gold = [], []

for _, r in df.iterrows():
    text = r["aspect"] + " [SEP] " + r["sentence"]
    inp = tokenizer(text, return_tensors="pt", truncation=True).to(DEVICE)
    with torch.no_grad():
        logits = model(**inp).logits
    preds.append(torch.argmax(logits).item())
    gold.append(label_map[r["sentiment"].lower()])

print(classification_report(gold, preds, target_names=["neg","neu","pos"]))
```

eval_ate:

```
import pandas as pd
import torch
from transformers import AutoTokenizer, AutoModelForTokenClassification
from peft import PeftModel
from seqeval.metrics import classification_report

DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
MODEL = "microsoft/deberta-v3-base"
```

```
tokenizer = AutoTokenizer.from_pretrained("ate_lora")
base = AutoModelForTokenClassification.from_pretrained(MODEL, num_labels=3)
model = PeftModel.from_pretrained(base, "ate_lora").to(DEVICE)
model.eval()

df = pd.read_csv("ate_test_final.csv")    # your final clean ATE dataset

preds, gold = [], []

for _, r in df.iterrows():
    words = r["sentence"].split()
    original_gold_tags = r["tags"].split() # Store original gold tags

    enc = tokenizer(
        words,
        is_split_into_words=True,
        truncation=True,
        padding="max_length",
        max_length=160,
        return_tensors="pt"
    ).to(DEVICE)

    with torch.no_grad():
        logits = model(**enc).logits

    pred_ids = torch.argmax(logits, dim=-1)[0].cpu().tolist()
    word_ids = enc.word_ids()

    pred_tags = []
    aligned_gold_tags = [] # New list for aligned gold tags
```

```
    for i, w in enumerate(word_ids):
        if w is not None:
            # Append predicted tag
            pred_tags.append(["O","B-ASP","I-ASP"][pred_ids[i]])
            # Append corresponding gold tag from original_gold_tags
            aligned_gold_tags.append(original_gold_tags[w])

    preds.append(pred_tags)
    gold.append(aligned_gold_tags) # Use aligned_gold_tags for evaluation

print(classification_report(gold, preds))
```

absa.py:

```
import os
import re
import torch
from peft import PeftModel
from transformers import (
    AutoTokenizer,
    AutoModelForTokenClassification,
    AutoModelForSequenceClassification
)

DEVICE = "cuda" if torch.cuda.is_available() else "cpu"

# -------- Globals (lazy-loaded) --------
tokenizer = None
ate_model = None
asc_model = None
```

```
def load_models():
    global tokenizer, ate_model, asc_model

    if tokenizer is not None:
        return  # already loaded

    BASE_DIR = os.path.dirname(os.path.abspath(__file__))
    PROJECT_ROOT = os.path.dirname(BASE_DIR)

    ATE_PATH = os.path.join(PROJECT_ROOT, "models", "ate_lora")
    ASC_PATH = os.path.join(PROJECT_ROOT, "models", "asc_lora")

    # ---- DeBERTa tokenizer ----
    tokenizer = AutoTokenizer.from_pretrained(
        "microsoft/deberta-v3-base",
        use_fast=True
    )

    # ---- ATE (token classification) ----
    ate_base = AutoModelForTokenClassification.from_pretrained(
        "microsoft/deberta-v3-base",
        num_labels=3,
        id2label={0: "O", 1: "B-ASP", 2: "I-ASP"},
        label2id={"O": 0, "B-ASP": 1, "I-ASP": 2},
        low_cpu_mem_usage=True
    )

    ate_model = PeftModel.from_pretrained(
        ate_base,
        ATE_PATH,
        local_files_only=True
```

```
    ).to(DEVICE)
    ate_model.eval()

    # ---- ASC (sequence classification) ----
    asc_base = AutoModelForSequenceClassification.from_pretrained(
        "microsoft/deberta-v3-base",
        num_labels=3,
        low_cpu_mem_usage=True
    )

    asc_model = PeftModel.from_pretrained(
        asc_base,
        ASC_PATH,
        local_files_only=True
    ).to(DEVICE)
    asc_model.eval()


# -------- ATE --------
def extract_aspects(text: str):
    enc = tokenizer(
        text,
        return_tensors="pt",
        truncation=True,
        max_length=512
    ).to(DEVICE)

    with torch.no_grad():
        logits = ate_model(**enc).logits

    preds = torch.argmax(logits, dim=-1)[0].cpu().tolist()
```

```
word_ids = enc.word_ids()

input_ids = enc["input_ids"][0].cpu().tolist()
tokens = tokenizer.convert_ids_to_tokens(input_ids)

aspects = []
current = []

for wid, label, tok in zip(word_ids, preds, tokens):
    if wid is None:
        continue

    clean_tok = tok.replace("▁", "").replace("##", "")

    if label == 1:  # B-ASP
        if current:
            aspects.append(" ".join(current))
        current = [clean_tok]

    elif label == 2 and current:  # I-ASP
        current.append(clean_tok)

    else:
        if current:
            aspects.append(" ".join(current))
            current = []

if current:
    aspects.append(" ".join(current))

return list(set(aspects))
```

```
# -------- ASC --------
def classify_sentiment(text: str, aspect: str):
    pair = f"{aspect} [SEP] {text}"

    inputs = tokenizer(
        pair,
        return_tensors="pt",
        truncation=True,
        max_length=512
    ).to(DEVICE)

    with torch.no_grad():
        logits = asc_model(**inputs).logits

    probs = torch.softmax(logits, dim=-1)
    conf, label = torch.max(probs, dim=-1)

    label_idx = label.item()
    confidence = int(conf.item() * 100) # Convert 0.95 -> 95

    sentiment = {0: "negative", 1: "neutral", 2: "positive"}[label_idx]
    return sentiment, confidence


# -------- MAIN ABSA PIPELINE --------
def run_absa(transcript: str):
    load_models()
```

```
    transcript = re.sub(r"\[.*?\]", "", transcript)

    aspects = extract_aspects(transcript)

    results = []
    for asp in aspects:
        sentiment, conf = classify_sentiment(transcript, asp)
        results.append({
            "aspect": asp,
            "sentiment": sentiment,
            "confidence": conf
        })

    return results
```

asr.py:

```
from faster_whisper import WhisperModel
import os
import dotenv
from groq import Groq

# -------------------- WHISPER SETUP --------------------
# Load once at startup — stays alive for the entire server lifetime

print("Loading Whisper model (small)...")
model = WhisperModel(
    "small",
    device="cpu",
    compute_type="int8"
)
```

```
print("Whisper model loaded.")

def get_model():
    return model


# -------------------- OPTIONAL LLM CLIENT --------------------

client = None
if os.getenv("GROQ_API_KEY"):
    client = Groq(api_key=os.getenv("GROQ_API_KEY"))


# -------------------- HELPERS --------------------

def format_time(sec: float) -> str:
    m = int(sec // 60)
    s = sec % 60
    return f"{m:02d}:{s:05.2f}"


def clean_asr_text(text: str) -> str:
    """
    Uses LLM to clean ASR output.
    This is OPTIONAL and only used when enabled from app.py
    """
    if client is None:
        return text   # fallback safely

    prompt = f"""
Correct transcription errors in the following ASR output.
```

```
Do NOT add new content or opinions.
Preserve meaning.

Text:
{text}
"""

    response = client.chat.completions.create(
        model="llama-3.3-70b-versatile",
        messages=[{"role": "user", "content": prompt}],
        temperature=0.1
    )

    return response.choices[0].message.content


# -------------------- MAIN TRANSCRIPTION --------------------

def transcribe(wav_path: str) -> str:
    """
    Transcribes audio into timestamped text.
    """
    segments, _ = get_model().transcribe(wav_path)

    output = []
    for s in segments:
        start = format_time(s.start)
        end = format_time(s.end)
        output.append(f"[{start}-{end}] {s.text.strip()}")

    return "\n".join(output)
```

Evaluation:

```
import pandas as pd
import torch
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from peft import PeftModel
from sklearn.metrics import classification_report

DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
MODEL = "microsoft/deberta-v3-base"

tokenizer = AutoTokenizer.from_pretrained("models/asc_lora")
base = AutoModelForSequenceClassification.from_pretrained(MODEL, num_labels=3)
model = PeftModel.from_pretrained(base, "models/asc_lora").to(DEVICE)
model.eval()

df = pd.read_csv("data/asc_test.csv")

label_map = {"negative":0, "neutral":1, "positive":2}
preds, gold = [], []

for _, r in df.iterrows():
    text = r["aspect"] + " [SEP] " + r["sentence"]
    inp = tokenizer(text, return_tensors="pt", truncation=True).to(DEVICE)
    with torch.no_grad():
        logits = model(**inp).logits
    preds.append(torch.argmax(logits).item())
    gold.append(label_map[r["sentiment"].lower()])

print(classification_report(gold, preds, target_names=["neg","neu","pos"]))
```